

BatchBase – Project Design Notes

1. Product Description

BatchBase will demonstrate its functionality using a mock pharmaceutical product.

The product is a **dry powder inhaler (DPI)** containing:

- 1 API (active pharmaceutical ingredient)
- 2 excipients:
 - EXP 1
 - EXP 2

2. Manufacturing Workflow

The manufacturing process for the mock product will follow this workflow:

1. Material Supply

A purchase order is created for the materials required for the in-process batch (blend batch).

The required materials are:

- API
- EXP 1
- EXP 2

Each material will have its own material/lot number and manufacturing date.

2. Blending Stage

The required materials are added to the blender:

- EXP 1
- API
- EXP 2

The materials are blended to produce the in-process blend batch.

The blend batch will have its own **in-process batch number**.

The start and end times of the blending process will be recorded.

3. Filling Stage

The blended powder is transferred to the filling process.

The powder is filled into individual pockets on a disc.

The filling process will have a recorded start and end time.

4. Assembly Stage

The filled disc is assembled into the final inhaler device.

At this stage, the final product is assigned a **finished product batch number**.

The finished product batch will therefore be linked back to the in-process blend batch from which it originated.

The assembly process will have a recorded start and end time.

3. QC Testing

The mock product will contain three main QC tests.

QC Test 1 – In-Process Blend Uniformity

This test is performed on the in-process blend.

- 10 samples are tested
- Each sample is assessed for potency
- Acceptance range: **90–110% potency**
- The individual sample results will be recorded
- The overall test will receive a pass/fail result

The test will be linked to the **in-process batch number**.

Future versions could include additional information such as:

- Equipment/machine used

- Analyst
- Test method
- Individual sample information

QC Test 2 – Finished Product Assay

This test is performed on the finished product.

- One replicate per batch
- The assay result is recorded
- Acceptance range: **95–110%**

The test will be linked to the **finished product batch number**.

Future versions could include additional information such as:

- Equipment/machine used
- Analyst
- Test method
- Individual sample information

QC Test 3 – Emitted Dose

This test is performed on the finished product.

- 20 discs are sampled across the batch
- The individual results are recorded
- Individual acceptance range 75 – 125%
- Mean acceptance range: **85–115%**
- RSD must be **less than 5%**

The test will be linked to the **finished product batch number**.

The database should ideally store the individual results as well as the calculated:

- Mean
- RSD
- Pass/fail result

Future versions could include additional information such as:

- Equipment/machine used

- Analyst
- Test method

4. Initial Database Tables

The initial version of BatchBase will use the following tables.

1. Raw Materials

Stores information about the raw materials used in manufacturing.

Potential information:

- Material/lot number
- Material name
- Manufacturing date
- Supplier/vendor

The mock project will assume that the materials are externally manufactured, so the manufacturing date can represent the vendor/manufacturer date.

2. Blend

Stores information about the in-process blending stage.

Potential information:

- In-process batch number
- Material(s) used in the batch
- Blend start date/time
- Blend end date/time

Future versions could include:

- Machine/equipment used
- Operators
- Manufacturing events
- Stoppages
- Additional process parameters

3. Fill

Stores information about the filling stage.

Potential information:

- In-process batch number
- Fill start date/time
- Fill end date/time

Future versions could include:

- Machine/equipment used
- Operators
- Manufacturing events
- Stoppages
- Additional process parameters

4. Assembly

Stores information about the final assembly stage.

Potential information:

- In-process batch number
- Finished product batch number
- Assembly start date/time
- Assembly end date/time

Future versions could include component information such as:

- Component batch/lot number
- Component expiry date
- Component type

5. Batch Crossover

This table will link the **in-process batch number** to the **finished product batch number**.

This relationship is important because it allows BatchBase to trace a finished product batch back through the manufacturing process to the original blend batch and raw materials.

6. QC Test 1 – Blend Uniformity

Stores the results of the in-process blend uniformity test.

Potential information:

- In-process batch number
- Test date
- Individual test results
- Mean/result where appropriate
- Pass/fail

Future versions could include:

- Machine/equipment
- Analyst
- Test method

7. QC Test 2 – Finished Product Assay

Stores the results of the finished product assay.

Potential information:

- Finished product batch number
- Test date
- Result
- Pass/fail

Future versions could include:

- Machine/equipment
- Analyst
- Test method

8. QC Test 3 – Finished Product Uniformity

Stores the results of the finished product uniformity test.

Potential information:

- Finished product batch number
- Test date
- Individual results
- Mean
- RSD
- Pass/fail

Future versions could include:

- Machine/equipment
- Analyst
- Test method

5. Future Development

The initial version will deliberately keep the data model relatively simple.

Future versions could introduce additional manufacturing information such as:

- Equipment and machine information
- Operators
- Manufacturing stoppages
- Process parameters
- Component traceability
- Additional QC tests
- More detailed laboratory results
- Investigation/event records
- Additional batch relationships

The aim of V1 is to demonstrate the core concept of **bringing related manufacturing, material and QC information together so that a batch investigation can be performed from one application.**